

DEVELOPER ARCHITECTURE MANUAL

Seekoji SMS Technical Deep-Dive

Complete Architecture, Finite State Machine Specifications, Database Schemas, API Layer & Frontend Design Tokens

Backend Infrastructure:

Laravel 12.0 (PHP 8.2+) / Eloquent ORM / Sanctum

State Machine Core:

App\Services\JobWorkflow FSM (8 Stages)

Frontend Architecture:

React 19 / Inertia.js 2.0 / Vite 7 / TailwindCSS 4

Automated Test Coverage:

PHPUnit Feature & Unit Suite (30 Passed, 116 Assertions)

1. Technology Stack & Architectural Principles

Seekoji SMS is designed as a high-performance modern monolithic web application. It leverages **Inertia.js** to combine the developer productivity of Laravel 12 with the rich reactivity of React 19 without building or maintaining a separate REST GraphQL layer.

COMPONENT	FRAMEWORK / LIBRARY	ARCHITECTURAL PURPOSE & IMPLEMENTATION DETAILS
Backend Core	Laravel 12.x (PHP 8.2+)	Handles routing, HTTP middleware gates, database transactions, request validation, and CSRF protection.
Monolith Adapter	Inertia.js 2.0	Renders server-side responses as client-side page props, eliminating REST API serialization boilerplate for page navigation.
Frontend Core	React 19	Modular component tree, local state hooks, custom drawer components, and reactive UI updates.
State Engine	App\Services\JobWorkflow	Finite State Machine (FSM) controlling all 8 stage transitions, outcome updates, and financial rule evaluations.
Sequence Service	App\Services\CounterService	Provides thread-safe atomic sequence generation for work order tokens and customer IDs via database row-level locks.
Build Engine	Vite 7.3 + TailwindCSS 4.0	Hot Module Replacement (HMR) development server and minified production bundles with CSS custom variables.
Database Engine	MySQL 8.0 / SQLite	Relational storage supporting transactions, JSON columns, soft/hard deletes, and indexed search queries.

2. Repository Directory Structure

The codebase strictly follows standard Laravel-Inertia conventions with dedicated service classes for complex business logic.

```
seekojisms/
├── app/
│   ├── Http/
│   │   ├── Controllers/
│   │   │   ├── Api/V1/
│   │   │   │   ├── CustomerController.php    # Customer API & hard delete handler
│   │   │   │   └── JobController.php          # Job CRUD, transition & token preview API
│   │   │   └── Web/
│   │   │       └── WebController.php          # Inertia page controllers
│   │   ├── Requests/
│   │   │   ├── UserStoreRequest.php          # User creation validation (password without confirmation)
│   │   │   ├── UserUpdateRequest.php         # User update validation
│   │   │   ├── JobStoreRequest.php           # Multi-product intake validation
│   │   │   └── JobTransitionRequest.php       # Stage transition payload rules
│   │   └── Resources/
│   │       ├── JobResource.php               # API transformer for jobs
│   │       └── CustomerResource.php           # API transformer for customers
│   ├── Models/
│   │   ├── Job.php                          # Job entity with stage/outcome casts & scopes
│   │   ├── Customer.php                     # Customer entity with customer_code cast
│   │   ├── JobEvent.php                     # Audit log entity
│   │   └── Counter.php                      # Sequence counter entity
│   └── Services/
```

			JobWorkflow.php	# Finite State Machine transition registry
			CounterService.php	# Atomic token generator & preview helper
			database/	
			migrations/	# Schema definitions for jobs, customers, job_events, counters
			seeders/DemoDataSeeder.php	# Initial seeders for development
			resources/	
			css/app.css	# Custom CSS design system, sk-tok, sk-btn, sk-card
			js/	
			Components/	
			Common/	# SearchableCustomerSelect, ConfirmActionModal
			Customers/	# CreateCustomerModal, ShowCustomerDrawer
			Jobs/	# CreateJobModal, JobTransitionModal, EditJobModal
			Pages/	
			Jobs/ControlCenter.jsx	# JCC split workspace page
			Jobs/Index.jsx	# All Jobs drawer & list workspace
			Customers/Index.jsx	# Customer directory & delete modal
			Accounts/Index.jsx	# Accounts settlement dashboard
			Reports/Index.jsx	# Reports & CSV/Print exports
			Users/Index.jsx	# Staff account management
			tests/	
			Feature/	
			CustomerTest.php	# Customer duplicate check & admin delete tests
			JobWorkflowTest.php	# FSM stage transition & multi-intake test suite

3. Database Schema & Entity Relationships

3.1 Entity Relationship Diagram & Foreign Keys

The core relational model revolves around `customers` , `jobs` , and `job_events` . Multi-product intake entries are grouped under a shared parent reference.

TABLE NAME	PRIMARY KEY	FOREIGN KEY RELATIONSHIPS	KEY INDEXED COLUMNS
<code>users</code>	<code>id</code> (BIGINT)	Belongs to Roles (Spatie Permission)	<code>email</code> (UNIQUE)
<code>customers</code>	<code>id</code> (BIGINT)	Has Many <code>jobs</code>	<code>customer_code</code> (UNIQUE), <code>mobile</code> (INDEX)
<code>jobs</code>	<code>id</code> (BIGINT)	Belongs to <code>customer_id</code> , Belongs to <code>technician_id</code> (Users), Belongs to <code>created_by</code> (Users)	<code>token_no</code> (UNIQUE), <code>stage</code> (INDEX), <code>outcome</code> (INDEX)
<code>job_events</code>	<code>id</code> (BIGINT)	Belongs to <code>job_id</code> (Jobs), Belongs to <code>user_id</code> (Users)	<code>job_id</code> , <code>created_at</code>
<code>counters</code>	<code>key</code> (VARCHAR)	None (Atomic counter lookup)	<code>key</code> (PRIMARY)

3.2 Detailed Field Specifications: jobs Table

COLUMN NAME	DATA TYPE & ATTRIBUTES	NULLABILITY	DESCRIPTION & BUSINESS RULES
token_no	VARCHAR(50)	NOT NULL	Auto-generated token string (e.g. SES1). Format configured via Settings.
product_name	VARCHAR(255)	NOT NULL	Equipment title (e.g. ACGT, VFD Drive 45kW).
brand	VARCHAR(255)	NULLABLE	Make/Brand manufacturer (e.g. ABB, Siemens, LG, Schneider).
serial_no	VARCHAR(255)	NULLABLE	Hardware serial identification number.
power_rating	VARCHAR(100)	NULLABLE	Power specification (e.g. 15 HP, 415V, 50Hz).
stage	ENUM(...)	NOT NULL	Current stage: new , testing , approval , repair , pending , completed , ready , delivered .
outcome	ENUM(...)	NULLABLE	Diagnostic/repair status: ok_no_fault , not_repairable , not_approved , work_done , cancelled .
estimated_budget	DECIMAL(10,2)	NULLABLE	Budget estimated during testing for customer approval.
approved_amount	DECIMAL(10,2)	NULLABLE	Customer-approved repair budget cap.
final_amount	DECIMAL(10,2)	NULLABLE	Final repair invoice total entered by technician at work completion.
payable_amount	DECIMAL(10,2)	NULLABLE	Actual receivable amount. Set to 0.00 when estimate is rejected or 0 fee.
is_paid	BOOLEAN	DEFAULT FALSE	Settlement flag. True when payment is recorded or marked zero due.

4. Finite State Machine (FSM) Core Specifications

The central business logic engine of Seekoji SMS is implemented in `App\Services\JobWorkflow`. This service acts as a deterministic state machine.

Transition Execution Pipeline: When `JobWorkflow::transition($job, $action, $data, $user)` is called:

1. Validates if `$action` is registered under current stage `$job->stage`. Throws HTTP 409 Conflict if illegal.
2. Executes required data validations (e.g., `estimated_budget` required for `fault_found`). Throws HTTP 422 if payload invalid.
3. Invokes state mutator callback to update `$job->stage`, `$job->outcome`, and financial snapshot fields inside a DB transaction.
4. Records an immutable audit record into `job_events` with action name, note, actor ID, and payload.

4.1 Complete Transition Handler Registry

CURRENT STAGE	ACTION TRIGGER	TARGET STAGE	SET OUTCOME	FINANCIAL & FIELD RULES
NEW	assign_tester	TESTING	None	Assigns <code>tester_id</code> . Records note.
NEW	cancel	COMPLETED	cancelled	Sets <code>payable_amount = 0</code> . Job terminated at intake.
TESTING	ok_no_fault	COMPLETED	ok_no_fault	Sets <code>payable_amount = inspection_fee</code> (or 0).
TESTING	fault_found	APPROVAL	None	Requires <code>estimated_budget > 0</code> . Stores <code>tester_findings</code> .
TESTING	not_repairable	COMPLETED	not_repairable	Sets <code>payable_amount = inspection_fee</code> . Stores findings.
APPROVAL	approve	REPAIR	None	Requires <code>approved_amount</code> and <code>technician_id</code> .
APPROVAL	not_approved	COMPLETED	not_approved	Sets <code>payable_amount = inspection_fee</code> (0 if waived). UI shows NO AMOUNT DUE.
REPAIR	work_done	COMPLETED	work_done	Requires <code>final_amount</code> . Sets <code>payable_amount = final_amount</code> .
REPAIR	hold	PENDING	None	Stores <code>pend_reason</code> (waiting for parts).
PENDING	resume	REPAIR	None	Resumes active repair work.
COMPLETED	collect_payment	READY	None	Sets <code>is_paid = true</code> , records <code>payment_mode</code> & <code>paid_at</code> .
COMPLETED	release_unpaid	READY	None	Leaves <code>is_paid = false</code> . Moves to ready for delivery with dues.
READY	deliver	DELIVERED	None	Requires <code>received_by</code> . Terminal stage transition.

5. Atomic Token Generation & Counter Service

To guarantee gapless, race-condition safe sequence numbers across concurrent user sessions, Seekoji SMS utilizes an atomic row-locking mechanism implemented in `App\Services\CounterService`.

```
// App\Services\CounterService.php Implementation
public static function generateNextToken(): string
{
    return DB::transaction(function () {
        $prefix = Setting::get('token_prefix', 'SES');
        $counter = Counter::where('key', 'job_token')->lockForUpdate()->first();

        if (!$counter) {
            $counter = Counter::create(['key' => 'job_token', 'value' => 0]);
        }

        $counter->value += 1;
        $counter->save();

        return $prefix . $counter->value;
    });
}
```

5.1 Pre-Generated Token Preview API

When users open the Intake Modal, the system calls `GET /api/v1/jobs/token-preview?count=N`.

- The helper `previewJobTokenInfo($count)` inspects current counter value `V` without incrementing database records.
- Returns preview list `["SES" + (V+1), "SES" + (V+2), ...]`.
- If the user closes the modal without submitting, zero counter IDs are wasted.

6. API Route Registry & Validation Specs

HTTP METHOD	ROUTE ENDPOINT	REQUEST CLASS / MIDDLEWARE	RESPONSE PAYLOAD / ACTION
GET	/api/v1/jobs/token-preview	auth:sanctum	Returns JSON <code>{ next_token: "SES1", range: "SES1 - SES3" }</code> .
POST	/api/v1/jobs	JobStoreRequest	Validates customer & product array. Creates jobs in DB transaction.
POST	/api/v1/jobs/{job}/transition	JobTransitionRequest	Executes FSM transition action, updates state & audit event log.
DELETE	/api/v1/jobs/{job}	role:admin	Executes <code>\$job->forceDelete()</code> to permanently purge database record.
DELETE	/api/v1/customers/{customer}	role:admin	Executes <code>\$customer->forceDelete()</code> for customer purging.

7. Testing Strategy & Quality Assurance

Seekoji SMS enforces strict unit and feature test coverage using PHPUnit. Every stage transition, permission gate, and counter generation logic is backed by automated tests.

```
# Execute Full PHPUnit Test Suite
php artisan test

# Test Output Matrix (30 Tests Passed, 116 Assertions)
PASS  Tests\Unit\ExampleTest
    ✓ that true is true

PASS  Tests\Feature\AuthAndPermissionsTest
    ✓ intake coordinator cannot call general transition endpoint
    ✓ intake coordinator can call deliver endpoint
    ✓ disabled user cannot authenticate
    ✓ cannot deactivate last active admin
    ✓ non admin cannot access accounts module

PASS  Tests\Feature\CustomerTest
    ✓ duplicate mobile number returns 422 with existing customer
    ✓ check mobile endpoint returns availability
    ✓ customer code is automatically generated
    ✓ only admin can update customer details
    ✓ only admin can toggle customer status
    ✓ only admin can delete customer

PASS  Tests\Feature\JobWorkflowTest
    ✓ new to testing via assign tester
    ✓ testing to approval via fault found
    ✓ approval to repair via approve
    ✓ repair to completed via work done
    ✓ pending to repair round trip
    ✓ invalid stage transition returns 409
    ✓ fault found without budget returns 422
    ✓ approve without technician returns 422
    ✓ deliver without delivery mode returns 422
    ✓ delivered stage is terminal and rejects further actions
    ✓ inspection fee is snapshotted on not repairable outcome
```


8. Production VPS Deployment & Operations Guide

Production deployment, queue worker management, and server administration guidelines for Seekoji SMS host environment.

8.1 VPS Server Environment Credentials

PARAMETER	CONFIGURATION VALUE	DESCRIPTION / USAGE
Server IP Address	72.61.236.77	Host VPS IP Address.
SSH Connection User	deploy	Primary deployment SSH user. Command: <code>ssh deploy@72.61.236.77</code>
VPS Console Password	123456	Web console login password for <code>deploy</code> user.
Admin Mail & Password	eseekoji@gmail.com / Seekoji@123	System administrator email credential.
Production Site Root	/var/www/service.seekojielectric.com	Application root directory on VPS.
Database Name	seekoji	MySQL Production Database.
Database Credentials	seekoji / Seekoji@123	MySQL database user and password.

8.2 Operational & Management Command Reference

MANAGEMENT TASK	SHELL COMMAND LINE	OPERATIONAL DESCRIPTION
Connect to VPS	<code>ssh deploy@72.61.236.77</code>	Establishes secure SSH terminal session to VPS.
Navigate to Site Path	<code>cd /var/www/service.seekojielectric.com</code>	Changes working directory to application root.
Deploy New Code	<code>./deploy.sh</code>	Pulls latest repository changes, builds, and deploys application.
Watch Live Logs	<code>tail -f storage/logs/laravel.log</code>	Streams real-time application error logs.
Queue Worker Status	<code>sudo supervisorctl status</code>	Inspects Supervisor background job queue status.
Restart Queue Worker	<code>sudo supervisorctl restart seekoji-worker:*</code>	Restarts all background queue worker instances.
MySQL Console	<code>mysql -u seekoji -p</code>	Opens production MySQL database terminal.
Restart PHP-FPM	<code>sudo systemctl restart php8.3-fpm</code>	Restarts PHP 8.3 FPM process manager.

